

CARLETON UNIVERSITY

FINAL EXAMINATION
MATH 2004 A
2019

DURATION: 3 HOURS

Department Name and Course Number: School of Mathematics and Statistics, MATH 2004 A
Course Coordinator: Professor A.B. Mingarelli

AUTHORIZED MEMORANDA
SCIENTIFIC CALCULATOR ONLY, NO WIRELESS DEVICES

In addition to the examination paper students will require an EXAMINATION BOOKLET, and a SCANTRON SHEET. This exam may be released to the Library.

1. Please verify that you are in possession of a SCANTRON FORM
2. Please fill in your COURSE CODE (e.g., BIT 2005 A), YOUR NAME and YOUR STUDENT NUMBER where required on the Scantron form AND on this examination.
3. **The entire examination consists of 6 pages and two parts, A and B, and is marked out of a total of 80, that is, 40% of your final mark. Part A consists of 12 multiple choice questions each worth 3 marks. Please fill in only one answer on your Scantron sheets with a pencil as there is only one answer to any given question. Circling two or more answers to any question invalidates that question (i.e., you get 0 marks for that question). Part B consists of 6 traditional type questions (with explanations required), for a total of 44 marks for that section. Both Part A and B must be submitted along with the scantron sheet, so do not detach nor unstaple this examination. DO NOT SUBMIT ROUGH WORK FOR PARTS A OR B. DO NOT SUBMIT EXAMINATION BOOKLET.**

Print Name :

Student Number:

Deferred Exam (Circle one): YES NO

- **A1.** Find the equation of the plane Π_1 through the two points $P(0, 1, 2)$ and $Q(-1, 0, 3)$ that is perpendicular to the plane Π_2 given by $2x - 4y + 6z = 2$.

(a) $-2x + 3y + z = 5$ (b) $2x - y + z = 1$ (c) $-x + 4y + 3z = 10$ (d) $-x + y = 1$ (e) None of these

Solution: (c)

- **A2.** Find the area of the region $R = \{(r, \theta) : 0 \leq r \leq 1, 0 \leq \theta \leq \pi/5\}$.

(a) $\frac{\pi}{10}$ (b) $\frac{2\pi}{15}$ (c) $\frac{5\pi}{3}$ (d) $\frac{\pi}{5}$ (e) None of these

Solution: (a)

- **A3.** Find the area of the conic section whose equation is $9x^2 + 4y^2 - 18x - 8y - 23 = 0$.

(a) 2π (b) 6π (c) 36π (d) 12π (e) None of these

Solution: (b)

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- **A4.** Which of the following points is on the straight line containing the two points $(2, 1, 3)$ and $(1, 2, -2)$?
 (a) $(-1, 4, 5)$ (b) $(-2, 1, -17)$ (c) $(1, 0, 1)$ (d) $(0, 2, -7)$ (e) None of these

Solution: (d)

- **A5.** Let $f(x, y)$ be given where $x = u(s, t), y = v(s, t)$. Assume further that $f_x = 2$ and $f_y = 3$ when $s = 1, t = 0$ and that $x_s = -1, y_s = 1$ when $s = 1, t = 0$. Find the value of $\frac{\partial f}{\partial s}$ when $s = 1, t = 0$.
 (a) 1 (b) -0 (c) 5 (d) -1 (e) None of these

Solution: (a)

- **A6.** Which of the following vector fields is conservative in $\mathbf{R}^2$?
 (a) $\mathbf{F}(x, y) = (x + e^y, y + e^{xy})$ (b) $\mathbf{F}(x, y, z) = (ze^{xz} \sin y, e^{xz} \cos y, xe^{xz} \sin y)$ (c) $\mathbf{F}(x, y) = (x^2 y^2 + 1, x^2 y^2 + 2)$
 (d) $\mathbf{F}(x, y, z) = (z, y, x^2)$ (e) None of these

Solution: (b)

- **A7.** The mass density $\delta(x, y)$, in kg/m of a linear cable of length L with initial point at $(0, 0)$ and endpoint at (L, L) is given by $\delta(x, y) = xy$. Find the mass of the cable.

- (a) $\frac{L^3}{3}$ (b) $\frac{\sqrt{2}L^2}{2}$ (c) $\frac{\sqrt{2}L^3}{3}$ (d) $\frac{2L^3}{3}$ (e) None of these

Solution: (c)

- **A8.** Which one of the following iterated integrals is equal to $\int_0^1 \int_x^1 e^{y^2} dy dx$?

- (a) $\int_0^1 \int_y^1 e^{y^2} dx dy$ (b) $\int_0^1 \int_0^y x e^{y^2} dx dy$ (c) $\int_0^1 \int_y^1 y e^{y^2} dx dy$ (d) $\int_0^1 \int_0^y e^{y^2} dx dy$
 (e) None of these

Solution: (d)

- **A9.** Find the volume of the solid region inside the paraboloid $z = x^2 + y^2$ that is cut off by the plane $z = 4$.

- (a) 8π (b) $\frac{81\pi}{2}$ (c) 16π (d) 4π (e) None of these

Solution: (a)

- **A10.** Find the smallest distance between the origin $(0, 0)$ and the hyperbola $xy = 1$ where $x > 0, y > 0$. (Hint: Use Lagrange multipliers.)
 (a) 4, (b) 8, (c) 1, (d) 2 (e) None of these

Solution: (c)

- **A11.** Evaluate the line integral $\int_C (x^2 - y) dx + (y^2 + x) dy$ where C is that part of the parabolic arc from $(0, 0)$ to $(1, 1)$.

- (a) 2 (b) -1 (c) 0 (d) 1 (e) None of these

Solution: (d)

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- **A12.** Let $\mathbf{F}$ be a vector field with $\operatorname{div} \mathbf{F}(x, y, z) = 8xyz$. Find the net flux through the closed surface $\mathcal{S} = \{(x, y, z) : 0 \leq x \leq a, 0 \leq y \leq b, 0 \leq z \leq c\}$

(a) ab^2c^2 (b) $a^2b^2c^2$ (c) a^2bc^2 (d) a^2b^2c (e) None of these

Solution: (b)

End of Part A

PART B: Do All six (6) Questions for a total of 44 marks out of a maximum of 80.

Do not detach nor unstaple this examination. Missing sheets will void any credit for those questions.

- **B1.** [8 marks] Evaluate $I = \int_C -y \, dx + x \, dy$ over the perimeter C of the closed curve $4x^2 + 9y^2 = 36$ oriented in a counterclockwise direction.

Solution: Using Green's Theorem we get

$$I = \iint_R 2 \, dA$$

where R is the elliptical region whose perimeter is C . Writing the ellipse in standard form gives $a = 3, b = 2$. Hence $I = 12\pi$ (or twice the area, πab , of the ellipse).

- **B2.** [7 marks] Evaluate the surface integral $I = \iint_{\mathcal{S}} y \, dS$ where $\mathcal{S}$ is the hemisphere $x^2 + y^2 + z^2 = R^2, z > 0$.

Solution: Observe that $\mathcal{S}$ is the union of the two half-hemispheres, $\mathcal{S}^+$, where $y > 0$, and $\mathcal{S}^-$ where $y < 0$. Each case is to be considered separately. Therefore,

$$I = I^+ + I^- = \iint_{\mathcal{S}^+} y \, dS + \iint_{\mathcal{S}^-} y \, dS.$$

Consider $\mathcal{S}^+$: Since $y > 0$, then $y = \sqrt{R^2 - x^2 - z^2} = g(x, z)$ defines half of $\mathcal{S}$. In this case,

$$I^+ = \iint_{\mathcal{R}^+} \sqrt{R^2 - x^2 - z^2} \sqrt{1 + g_x^2 + g_z^2} \, dA = \iint_{\mathcal{R}^+} \sqrt{R^2 - x^2 - z^2} \frac{R}{\sqrt{R^2 - x^2 - z^2}} \, dA = R \iint_{\mathcal{R}^+} dA$$

where $\mathcal{R}^+$ is the projection of $\mathcal{S}^+$ onto the xz -plane, i.e., $\mathcal{R}^+$ is the half disk in the xz -plane given by $\mathcal{R}^+ = \{(r, \theta) : 0 \leq r \leq R, 0 \leq \theta \leq \pi\}$ (in polar coordinates centered at the origin). The last integral is equal to the area of $\mathcal{R}^+$ which is, of course, $\pi R^2/2$. Thus,

$$I^+ = R \frac{\pi R^2}{2} = \frac{\pi R^3}{2}.$$

In the case of $\mathcal{S}^-$ we have $y < 0$. we have $y = -\sqrt{R^2 - x^2 - z^2}$ and so exactly the same calculation (with the negative sign attached) now gives

$$I^- = -R \frac{\pi R^2}{2} = -\frac{\pi R^3}{2}.$$

Adding these two values $I^+ + I^-$, we get

$$I = 0.$$

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- **B3.** [7 marks] Let $\mathbf{F}(x, y, z) = (-y^2, x, z^2)$ and let $\mathcal{S}$ be that part of the plane $x + z = 2$ cut by the cylinder $x^2 + y^2 = 1$. Use any method to evaluate the flux integral $I = \iint_{\mathcal{S}} (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS$ where $\mathbf{n}$ is an outward pointing unit normal vector to $\mathcal{S}$.

Solution: It is easy to see that $\text{curl } \mathbf{F} = (0, 0, 1 + 2y)$. In addition, since $\mathcal{S}$ is part of the plane $x + z = 2$, it follows that a normal vector is given by $\pm(1, 0, 1)$. The outward pointing unit normal is therefore $(1, 0, 1)/\sqrt{2}$. Thus, $(\text{curl } \mathbf{F}) \cdot \mathbf{n} = \frac{1 + 2y}{\sqrt{2}}$. On the other hand, the surface is $z = 2 - x$ so $dS = \sqrt{2} \, dA$ from which we get $(\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS = \frac{1 + 2y}{\sqrt{2}} \sqrt{2} \, dA = (1 + 2y) \, dA$. Therefore, $I = \iint_{\mathcal{S}} (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS = \iint_{\mathcal{R}} (1 + 2y) \, dA$ and projecting $\mathcal{S}$ onto the xy -plane we get that $\mathcal{R}$ is the disk $x^2 + y^2 \leq 1$. Using polar coordinates, we find

$$I = \iint_{\mathcal{R}} (1 + 2y) \, dA = \int_0^{2\pi} \int_0^1 (1 + 2r \sin \theta) r \, dr \, d\theta = \pi.$$

- **B4.** [7 marks] Use any method to evaluate the flux integral $\iint_{\mathcal{S}} \mathbf{F} \cdot \mathbf{n} \, dS$ of the vector field $\mathbf{F}(x, y, z) = (x, z^2x, y^3x^2)$ over that part of the closed hemisphere $x^2 + y^2 + z^2 = 4$, $z \geq 0$.

Solution: Use the Divergence Theorem. Since the surface is closed we see that $\text{div } \mathbf{F} = 1$ everywhere in the hemisphere. Denoting the closed solid hemisphere by T we get,

$$\iint_{\mathcal{S}} \mathbf{F} \cdot \mathbf{n} \, dS = \iiint_T \text{div } \mathbf{F} \, dV = \iiint_T 1 \, dV = \frac{16\pi}{3}$$

as the last integral is simply the volume of T .

- **B5.** [2+5 marks] Let $\mathbf{F}(x, y, z) = e^{y+z} \mathbf{i} + xe^{y+z} \mathbf{j} + xe^{y+z} \mathbf{k}$. Determine whether $\mathbf{F}$ is a gradient field and if so, find its potential function $f(x, y, z)$. If no such function f exists, then explain why not.

Solution: It is easy to see that the curl of $\mathbf{F}$ is zero. Hence $\mathbf{F}$ is a gradient field and there is a potential function f such that $\nabla f = \mathbf{F}$. The easiest way to determine f is by the method of line integrals. Observe that $\mathbf{F}$ is continuous everywhere on $\mathcal{C}$. Thus, $f(x, y, z) = \int_{\mathcal{C}} \mathbf{F} \cdot d\mathbf{r}$ where $\mathcal{C}$ is the straight line joining $(0, 0, 0)$ to an arbitrary point (x, y, z) . Parametrizing $\mathcal{C}$ we get $\mathbf{r}(t) = (tx, ty, tz)$ where $t \in [0, 1]$. Thus, $\int_{\mathcal{C}} \mathbf{F} \cdot d\mathbf{r} = \int_{\mathcal{C}} \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) \, dt = \int_{\mathcal{C}} \mathbf{F}(tx, ty, tz) \cdot (x, y, z) \, dt$ and so

$$\begin{aligned} f(x, y, z) &= \int_0^1 \mathbf{F}(tx, ty, tz) \cdot (x, y, z) \, dt \\ &= \int_0^1 (xe^{t(y+z)} + txye^{t(y+z)} + txyz e^{t(y+z)}) \, dt \\ &= \dots \\ &= xe^{y+z} + C \end{aligned}$$

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- **B6.** [8 marks] Find and classify the critical points of the function $f(x, y) = 8x^2 + 4y^2 + 4y + 5$ for $(x, y) \in \mathcal{D}$ where $\mathcal{D} = \{(x, y) : x^2 + y^2 \leq 1\}$.
(A total of 4 bonus marks is given for finding the absolute maximum and absolute minimum of f on $\mathcal{D}$ and giving reasons.)

Solution: The interior critical points are given by setting $f_x = 16x = 0$ only when $x = 0$, and $f_y = 8y + 4 = 0$ which holds only when $y = -1/2$. Thus, $(0, -1/2)$ is the only interior critical point.

Note that $f_{xx}f_{yy} - f_{xy}^2 = 16 \cdot 8 - 0 > 0$.

In addition, $f_{xx}(0, -1/2) = 16 > 0$ and so $(0, -1/2)$ is a local minimum. The minimum value is 4.

Critical points on the boundary are obtained by setting $x^2 = 1 - y^2$ into f thereby yielding a function $g(y) = -4y^2 + 4y + 13$, where $y \in [0, 1]$. The critical points are given by setting $g'(y) = 0$ or $-8y + 4 = 0$ which gives $y = 1/2$. Since $x^2 = 1 - y^2$ it follows that $(\pm\sqrt{3}/2, 1/2)$ are the only two critical points on the boundary. Since $g''(y) = -8 < 0$ it follows that these two points are local maxima. The maximum value here is 14.

So, the only local minimum is at $(0, -1/2)$ and the local maxima are at $(\pm\sqrt{3}/2, 1/2)$.

TOTAL : 80